

Element C

The group's design requirements came from both personal research as well as the conversations the group had with engineers from Drexel and the supervisor of the students we are designing for. Requirements 1-3 and 5-7 were all given to the group directly from Mrs. Brennan, however we were able to expand on them through our own research. The priority ranking goes from most important to least important, and the high priority are mostly requirements from Mrs. Brennan. However, by researching how each human neurodivergent brain works differently and how sensory objects impact them, as well as hearing about each student's strengths and skills, allowed us to make the product more universal.

Prioritized Design Requirement List:

High Priority

1. Have stimulating reward
 - a. Correct matches result in lights, sounds, textures, and/or vibrations
 - b. Must have at least 2 out of 4
2. Educational value
 - a. Students must be able to work on educational things that reflect what is being learned in the class
3. Adjustable, (buttons can be specialized for students)
 - a. Tiles/buttons are removable from base of design and can be switched out for different educational matchesroom
4. Safety
 - a. Edges are rounded
 - b. Pieces are mouth safe
 - c. No wires are exposed to the children
5. Mouth safe
 - a. Pieces big enough to not be swallowed
 - b. Material cannot be sullied easily
 - c. Easy to clean and sanitize
6. Mobile
 - a. Can be picked up and has a weight in between **insert weight here**
 - b. Easily transportable and storable

Lower Priority

7. Can be turned on and off by a button or switch
8. Buttons can still be pressed by students with motor disabilities
 - a. Buttons large enough to be pressed by a closed fist/hand
9. Pieces are secured to the device
 - a. Parts can't be easily removed by the students
10. Number of tiles
 - a. Lots of tiles and options for the students

11. Tests students motor skills

- a. Incorporates some element of turning or moving that requires

12. Durability

- a. Lasts with multiple plays each day
- b. Lasts for over a year in the classroom

13. Aesthetics

- a. Toy is built with engaging bright colors
- b. Also includes textures

High Priority	Design Requirements	Explanation	What is measured?
1	Have stimulating reward	The project is based on a cause and effect stimulation toy so it is the most important requirement. Correct matches of the buttons will result in some sort of stimulating reward. This includes flashing lights, buzzing noises, and vibrations.	Our requirements state that we must have AT LEAST two out of these four stimulants. However three out of four or four out of four would be preferred.
2	Educational value	Our stakeholder requested that this toy be of educational value, and use matching to enhance stimulation and development.	The matches on the toy help the students further develop what they are learning in class.
3	Adjustable, (buttons can be specialized for students)	Different students are at different educational and developmental stages, so if the buttons are customizable for students they will all be able to use them.	The tiles and buttons are removable from the base of design and can be switched out for different matches.
4	Safety	The most important part of the device itself is making sure it keeps the students safe and will not hurt them. It is an educational tool and we do not want to cause any injury as it is a mechanical device that could potentially be	There are no edges or pieces that can be choked on and the device cannot be easily thrown by the students. All the electronics are safely inside the device and there are no dangers to the kids.
5	Mouth safe	The kids often put things in their mouths so all the pieces must be big enough to not be swallowed, the material cannot be sullied easily, and it must be easy to clean and sanitize.	The tiles must be at least size of x by x to ensure it cannot be swallowed and the material of the device and pieces must be able to get wet and be cleaned without deteriorating.

6	Mobile	The device must be able to be picked up by any of the nurses, aids, or teachers. However it must also be heavy enough that if a student is using it they will not knock it over or be able to throw it. It also must fit in the storage spaces available.	The device cannot be more than 2 feet tall or wide, and it must be within the weight range of 5-10 lbs.
Low Priority	Design Requirements	Explanation	What is measured?
7	Can be turned on and off by a button or switch	Mrs. Brennan requested that the toy be operated by a small on/off button or switch. We don't want it somewhere that it can be accidentally hit by a kid using the device.	It must have a switch or a button which powers the arduino on and off. This button must be away from the buttons so it is not hit by a student by accident.
8	Buttons can still be pressed by students with motor disabilities	Some students have difficulty controlling their arms and hands, so if they have muscular disabilities we still want them to be able to hit the buttons. This can be done with larger buttons spaced farther apart.	The buttons must be the size of a fist and spaced about an inch apart so the students can hit the buttons even within a certain degree of precision.
9	Pieces are secured to the device	The pieces falling off the device when it is used would make it ineffectual and overall a worse experience for the students. However they still have to be removable.	The pieces are secured to the device and will not fall off if hit or knocked over.
10	Number of tiles	The toy must be engaging for a long period of time so the more matches there are, the longer kids will stay interested and engaged.	This is measured by the number of tiles on each board as well as how many specific sets of tiles we create to interchange with the board.
11	Tests students motor skills	The device already had educational value but by requiring more in depth use of fine motor skills like rotating or lifting would add another level of complexity to the design.	The tiles already require a degree of fine motor skills and another added layer could be spinning the device to see more matches.

12	Durability	The toy must be useful of course, but it also has to last. The group's youngest member is a Junior in high school meaning she will only be able to fix it for another year, therefore the toy has to last without constant testing and modification.	The device is built from a sturdy material like wood and is not easily breakable. All the parts must be secured and
13	Aesthetics	In order for the kids to stay engaged in the toy, it must be visually appealing. This includes possible textures on the build.	Are there 3 or more colors on the toy? Are there textures included in the build besides just smooth surfaces?

Functions:	Advanced	Proficient	Acceptable	Unacceptable
Must have stimulating reward	Device provides light, sound, and vibration as a the "reward"	Includes sounds and lights in "reward"	Includes light in "reward"	Device does not provide light, sound, or vibration as a "reward"
Mouth safe	Made of silicone. [Material used is non toxic and pieces are unable to fit inside the mouth]	Made of rubber. Made of 3D printing plastic and sanded down for safety and then sealed.	Made of 3D printing plastic with no extra safety precautions made.	Made of wood, especially painted wood. [Improper material is used that doesn't ensure a safe environment]
Adjustable (buttons can be specialized for students)	Buttons are adjustable, many different options of shape/size to switch out + extra spare pieces	Buttons are adjustable with different pieces to swap out as well at spare pieces	Buttons are adjustable with extra spare pieces	Buttons are not adjustable, there is no variety or spare pieces
Can be turned on and off by a button or switch	Device is powered on/off through an easily accessible on/off switch. Switch is covered for teacher access only.	Device is powered through an on/off switch, it's easily accessible and durable.	Device is powered through an on/off switch.	Device is not powered with an on/off switch.
Mobile	The device is lightweight with manageable construction. Device is secured and	The device is lightweight, it is consistently secured at all points while being very durable.	The device is within 5 lbs-10 lbs, it's adequately secure and durable.	The device is extremely cumbersome, significantly more than 10 lbs in weight

	durable at all points.			and fragile
Buttons can still be pressed by students with motor disabilities	Buttons are large and equally spaced with plenty of room in between to allow for accuracy.	Buttons are large enough, equally spaced and have space in between them.	Buttons are large and equally spaced from one another.	Buttons are small, close to one another, and unequally spaced - creating a difficult situation for children with motor disabilities.
Pieces are secured to the device	Pieces are secured to the device while also being interchangeable and able to be cleaned.	Pieces are secured to the device with little to no trouble switching or cleaning the tiles.	Pieces are secured to the device.	Pieces are unsecured, creating a faulty and unstable device.
Number of tiles	8-10 tiles	5-7 tiles	2-4 tiles	0-1 tiles

The group emailed the primary stakeholder Stephanie Brennan saying, “Hi Ms. Brennan, The group is finalizing design requirements, or just things our device must include. Below are the groups requirements ranked high and low, high being what WILL 100% be included in the final design, and low being what MAY be included in the final design. Please let us know if we are missing anything you think would be completely vital to the design (for high priority) or something you would like to be included in the "nice to have" section (low priority)” and then listing the design requirements. Ms. Brennan responded with “Looks good!”.